

Explain Clustering algorithms and their use cases

What is Clustering?

Clustering is an **unsupervised machine learning** technique used to group similar data points into **clusters** without having predefined labels. The goal is to maximize similarity **within clusters** and minimize similarity **between clusters**.

1. K-Means Clustering

How It Works:

- Partitions the dataset into **K** clusters.
- Assigns each point to the **nearest cluster center** (centroid).
- Updates centroids iteratively until convergence.

Use Cases:

- **Customer segmentation** (e.g., grouping customers by purchasing behavior)
- **Market basket analysis**
- **Image compression** (e.g., reducing colors in an image)
- **Anomaly detection**

2. Hierarchical Clustering

How It Works:

- Builds a tree-like structure (dendrogram) of clusters.
- Two types:
 - **Agglomerative (bottom-up)**: each point is a cluster and merged iteratively.
 - **Divisive (top-down)**: all points in one cluster, and split iteratively.

Use Cases:

- **Genetic data analysis** (e.g., phylogenetic trees)

- **Document or text clustering**
- **Social network analysis**
- **Grouping products by similarity**

DBSCAN (Density-Based Spatial Clustering of Applications with Noise)

How It Works:

- Groups data based on **density**.
- Can find arbitrarily shaped clusters.
- Identifies outliers as noise.

Use Cases:

- **Geospatial clustering** (e.g., identifying hotspots on a map)
- **Fraud detection**
- **Clustering images or time series with noise**
- **Social media trend analysis**